RIG-SURJECTIVITY OF F WITH RIG-ETALE RIG-SURJECTIVE DIAGONAL IS TESTED ON DVR POINTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof reduces rig-surjectivity to the flat-locus dichotomy, checks the diagonal hypothesis passes to the base change, and uses the factoring DVR extension to exclude the empty case

Editorial note. The label lemma-rig-monomorphism-rig-surjective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- rig-surjectivity of morphisms of formal algebraic spaces tested by adic morphisms from complete DVRs after extension
- base change preserves local finite type (Formal Spaces, Lemma formal-spaces-lemma-base-change-finite-type)
- the diagonal of a base change is the base change of the diagonal, preserving rig-etale and rig-surjective
- the dichotomy for the flat locus of a morphism to Spf(R) with rig-etale rig-surjective diagonal: empty or everything (lemma-flat-and-diagonal-rig-surjective of the same chapter)

Source reference. Original lemma 0AQ5: sources/stacks-chow/tags.txt, source lines 13498–13498. Prerequisite pointer invoked at proof line(s) 5931; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GD9: sources/stacks-restricted/pinned-original-source.tex, source lines 5832–5832. Prerequisite pointer invoked at proof line(s) 5934; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces. Assume*

(1) *X and Y are locally Noetherian,*
(2) *f locally of finite type,*
(3) *$\Delta_f : X \to X \times_Y X$ is rig-étale and rig-surjective.*

Then f is rig surjective if and only if every adic morphism $Spf(R) \to Y$ where R is a complete discrete valuation ring lifts to a morphism $Spf(R) \to X$.

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Proof. One direction is trivial. For the other, suppose that $\mathrm{Spf}(R) \to Y$ is an adic morphism such that there exists an extension of complete discrete valuation rings $R \subset R'$ with $\mathrm{Spf}(R') \to \mathrm{Spf}(R) \to X$ factoring through Y. Consider the fibre product diagram

$$\begin{array}{ccccc} \mathrm{Spf}(R') & \longrightarrow & \mathrm{Spf}(R) \times_Y X & \longrightarrow & X \\ & \searrow & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle f} \\ & & \mathrm{Spf}(R) & \longrightarrow & Y \end{array}$$

The morphism p is locally of finite type as a base change of f, see Formal Spaces, Lemma 0AQ5. The diagonal morphism Δ_p is the base change of Δ_f and hence is rig-étale and rig-surjective. By Lemma 0GD9 the flat locus of $\mathrm{Spf}(R) \times_Y X$ over R is either $\emptyset$ or equal to $\mathrm{Spf}(R)$. However, since $\mathrm{Spf}(R')$ factors through it we conclude it is not empty and hence we get a morphism $\mathrm{Spf}(R) \to \mathrm{Spf}(R) \times_Y X \to X$ as desired. $\square$